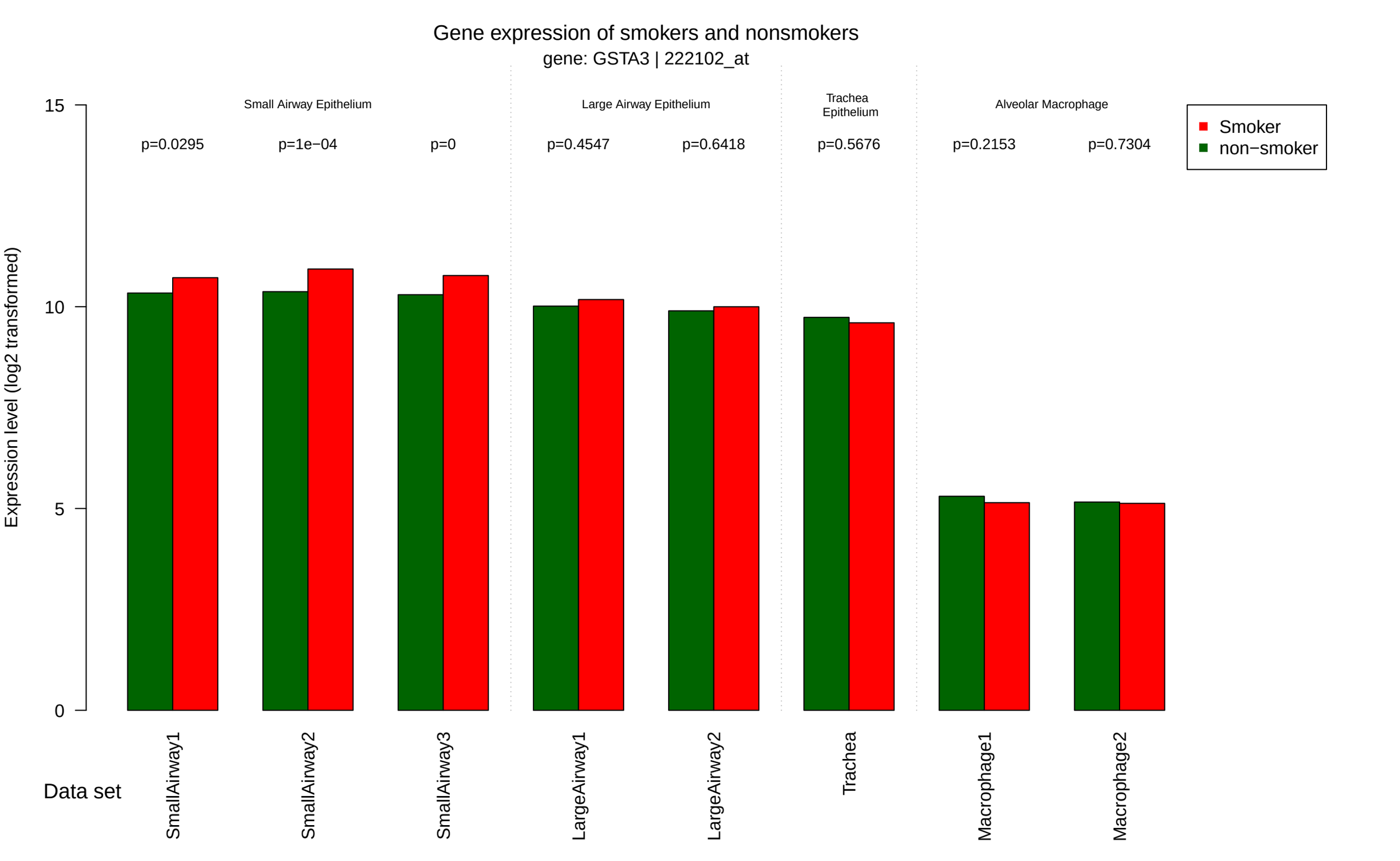
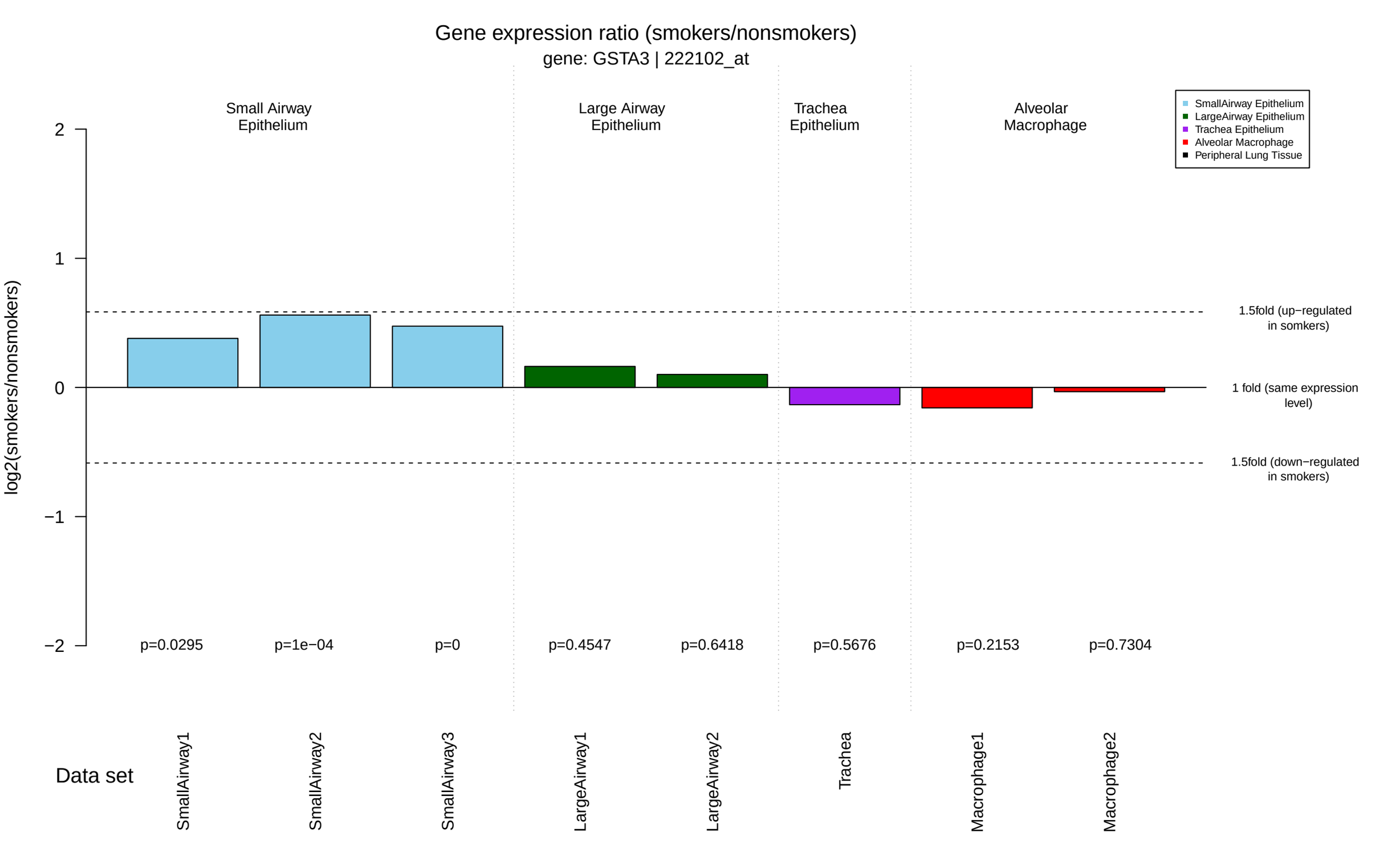
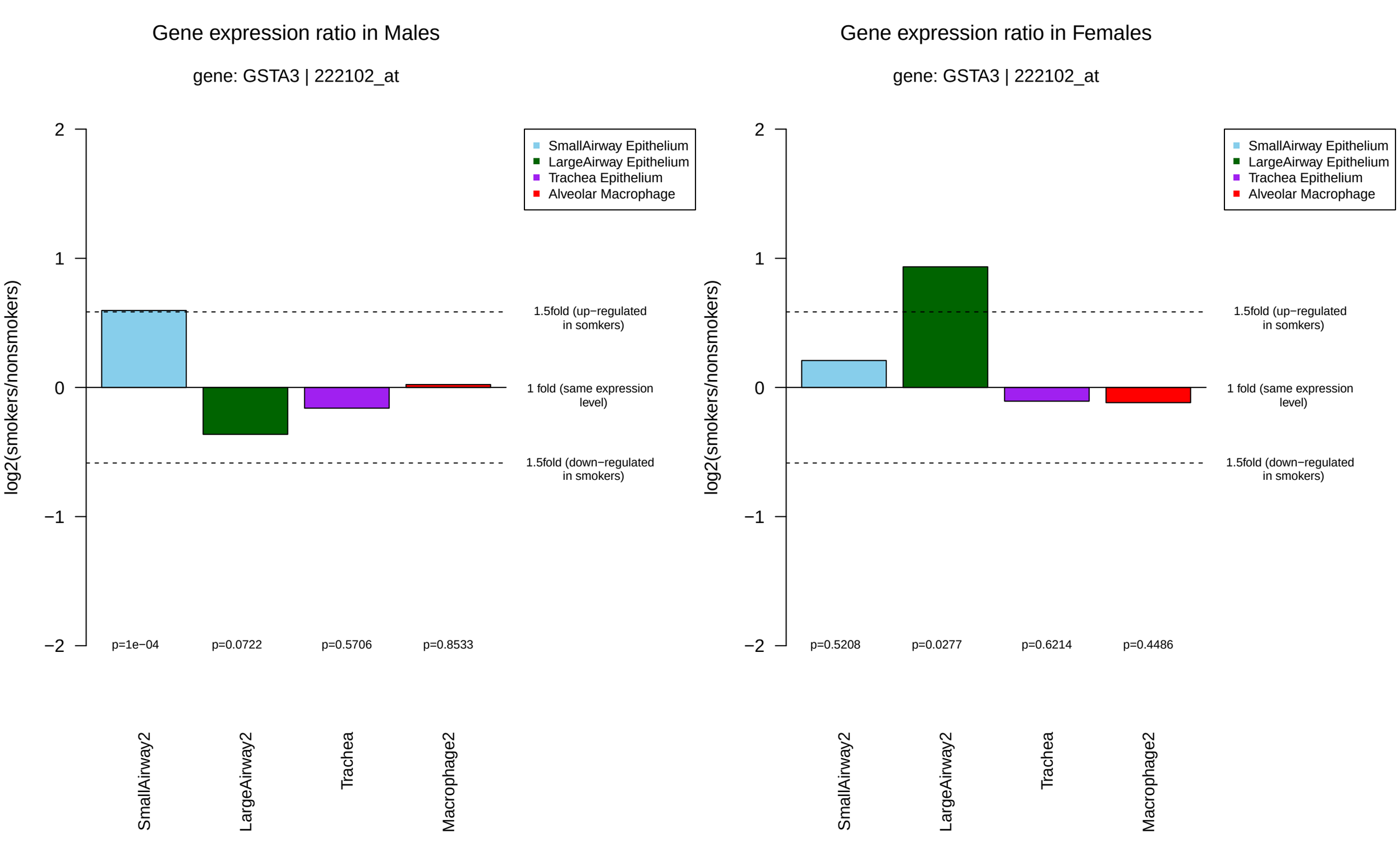
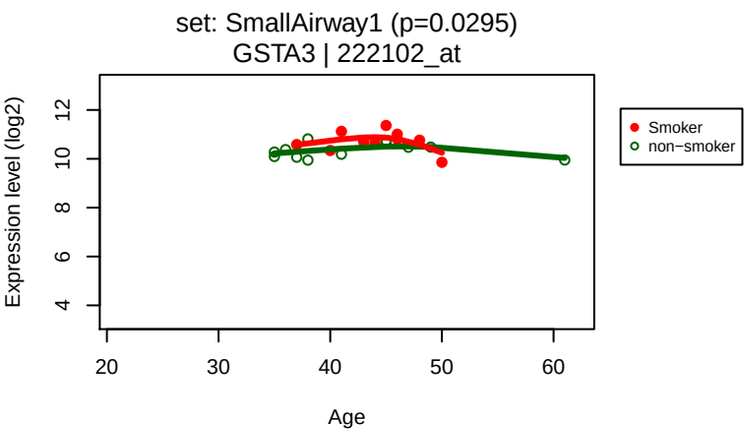
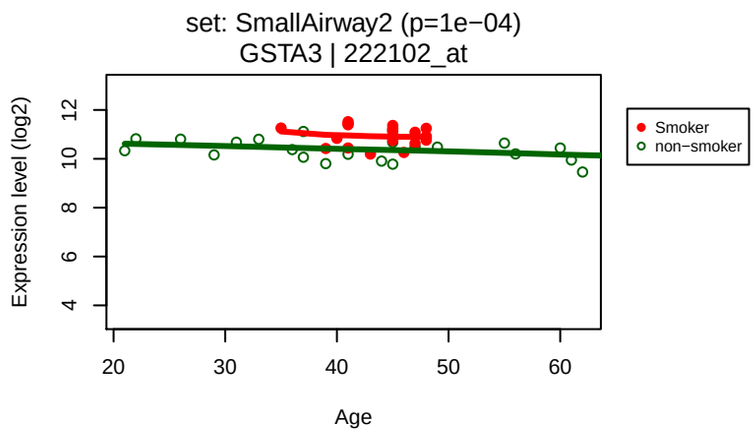
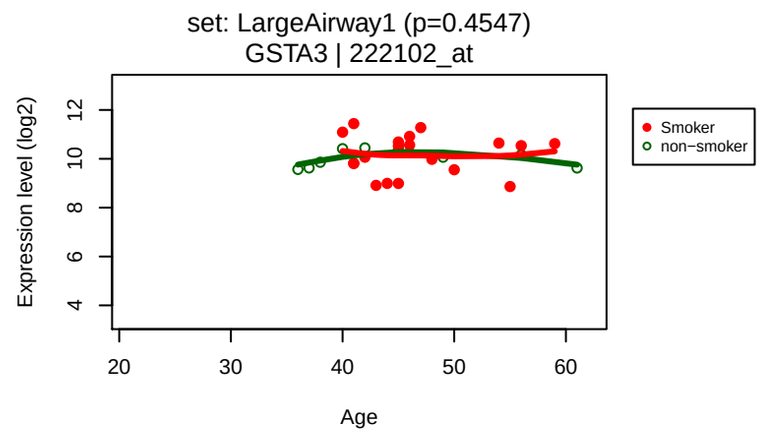

Gene expression of smokers and nonsmokers
gene: GSTA3 | 222102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0295
p=1e−04
p=0
p=0.4547
p=0.6418
p=0.5676
p=0.2153
p=0.7304
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

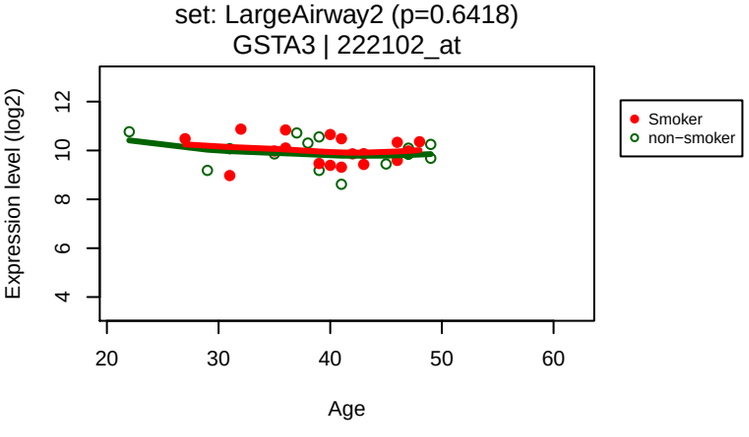

Gene expression ratio (smokers/nonsmokers)
gene: GSTA3 | 222102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0295
p=1e−04
p=0
p=0.4547
p=0.6418
p=0.5676
p=0.2153
p=0.7304
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

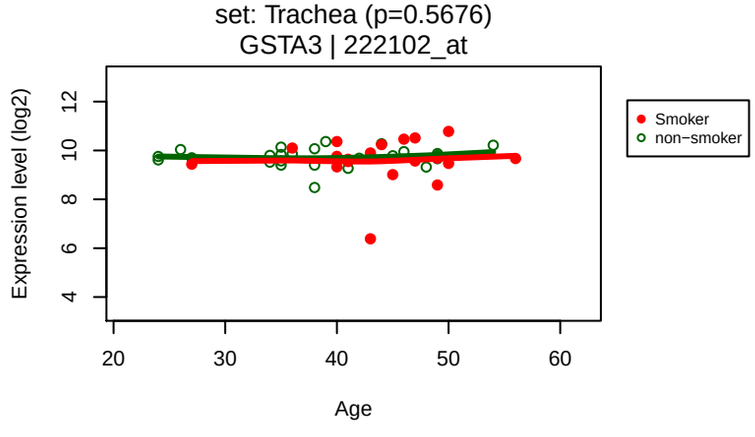

Gene expression ratio in Males
gene: GSTA3 | 222102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0722
p=0.5706
p=0.8533
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTA3 | 222102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5208
p=0.0277
p=0.6214
p=0.4486
SmallAirway2
LargeAirway2
Trachea
Macrophage2

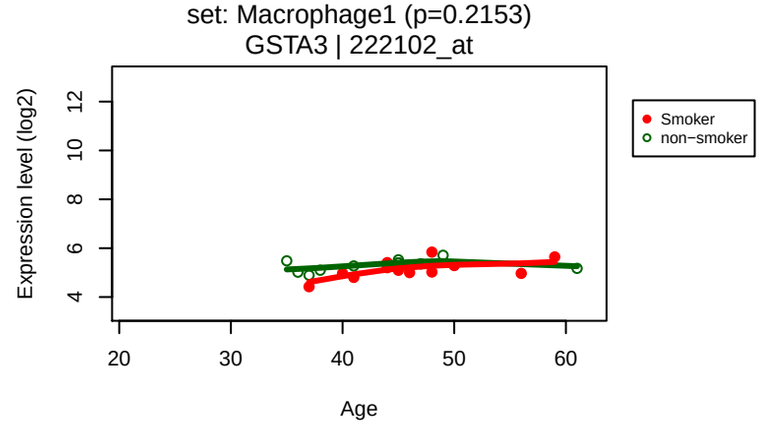

set: SmallAirway1 (p=0.0295)
GSTA3 | 222102_at
Expression level (log2)
Age
Smoker
non−smoker

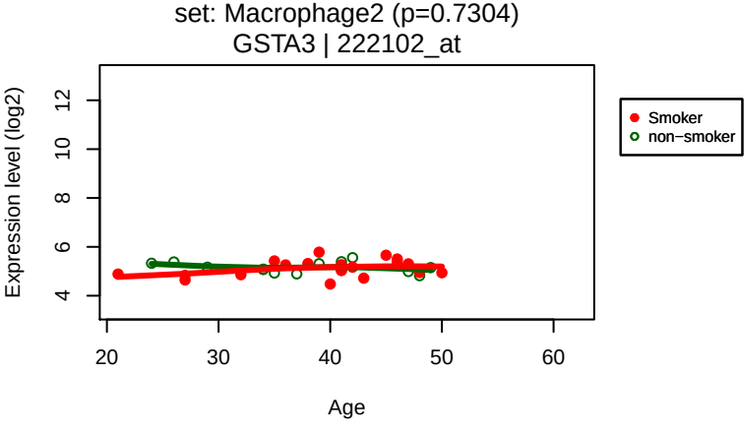

set: SmallAirway2 (p=1e−04)
GSTA3 | 222102_at
Expression level (log2)
Age
Smoker
non−smoker

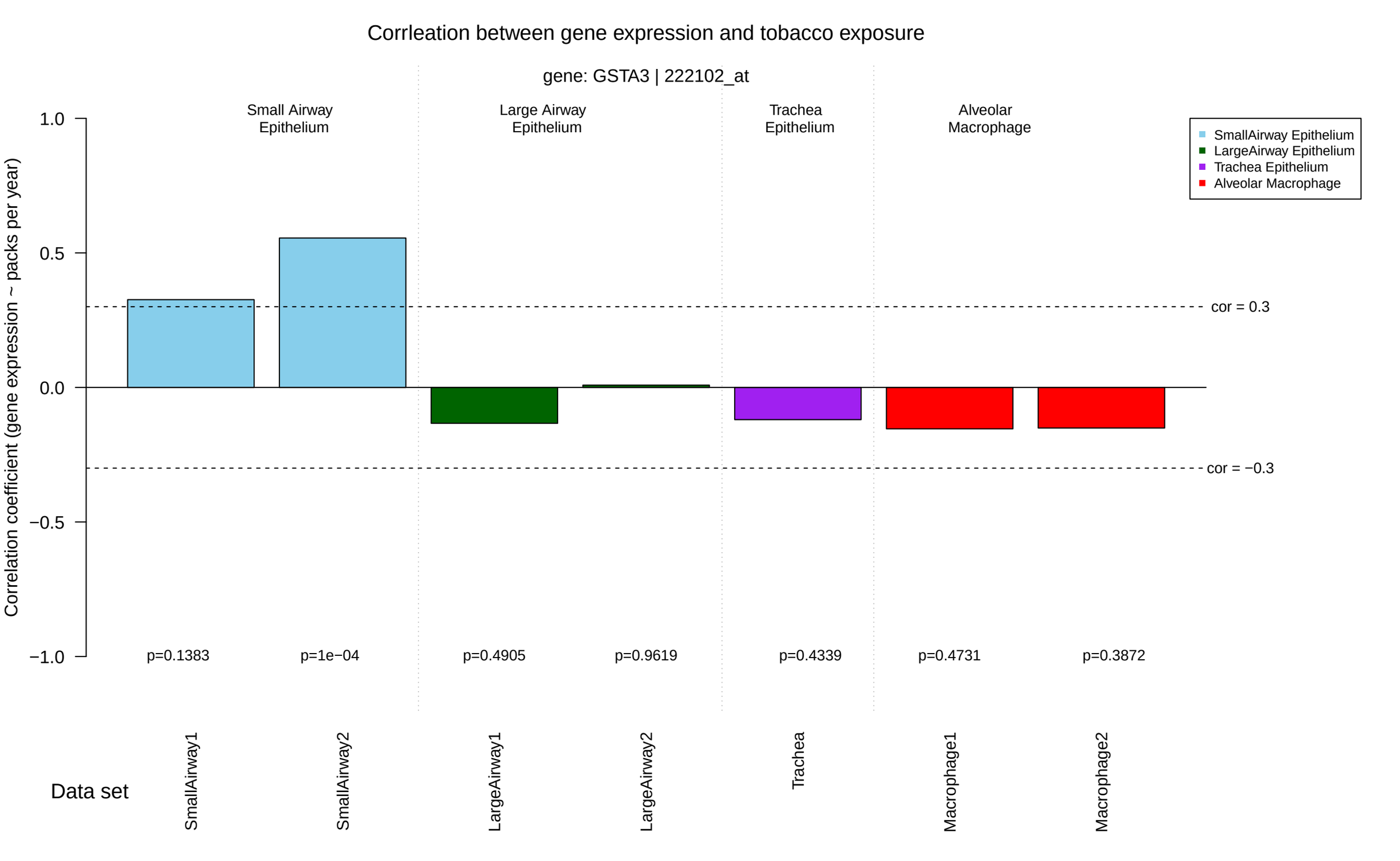

set: LargeAirway1 (p=0.4547)
GSTA3 | 222102_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6418)
GSTA3 | 222102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5676)
GSTA3 | 222102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2153)
GSTA3 | 222102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7304)
GSTA3 | 222102_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTA3 | 222102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1383
p=1e−04
p=0.4905
p=0.9619
p=0.4339
p=0.4731
p=0.3872
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2